Reference Event 1: The Aliso Canyon Gas Release, October 2015 to February 2016

Characterization Against the Natural Gas Sector Design Basis

DBA-GAS-RE-1 — Reference Event Instrument, DBA-GAS Series

Lineage: Issued under the Natural Gas Sector Framework v1.12; Energy Sector Framework; DBA-MA Parent Framework; conferred on the proposal of Underground Storage (DBA-GAS-FC3)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the uncontrolled release of natural gas from well Standard Sesnon 25 (SS-25) at the Southern California Gas Company's Aliso Canyon storage field, Los Angeles County, California, from 23 October 2015 until the well was sealed in February 2016, back through DBA-GAS-FC3 (*Underground Storage*), whose postulated event EF-1, storage well casing integrity failure and blowout, it characterizes (§3.2). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Natural Gas Sector Framework v1.12, which the class inherits together with the Energy Sector Framework (DBA-EN-SF) and the DBA-MA parent framework, and was conferred by the framework's §10.4 on the author's decision of 15 October 2026, on the proposal of *Underground Storage* at its §8.6. It is the natural gas sub-sector's first reference event instrument and the first in the energy series. Section references of the bare form §N are to *Underground Storage* v0.10, the version this instrument now reads, and the framework is read at v1.13. Version 0.1 read the class at v0.9, whose section numbers v0.10 keeps; the proposal was made at v0.8, and between v0.8 and v0.9 the class's section numbers for the blowout-contractor constraint (§7.2) and the federal recommendation (§8.2) changed.

The scope runs from the discovery of the leak on 23 October 2015, through the attempts to stop it and its sealing, to the field's limited return to service on 19 July 2017 and the inventory limits set since, the last of them on 31 August 2023. The technical record of the release is closed. The future of the field is before the California Public Utilities Commission in an investigation opened in 2017 into the feasibility of minimizing or eliminating its use, and in that respect the record is open.

No individual is named. The company, its parent Sempra, the state and federal agencies and the root-cause investigator are named as institutions. The company pleaded no contest in September 2016 to one misdemeanor count of failing to report the leak immediately to state officials, and that is stated as the court record states it. The regulators' and the root-cause investigator's reports are primary, and this instrument makes no finding of fault. No question of attribution arises: no source read suggests that any party caused the failure.

The well, the field and its surface works lie within this class (§1.2). Aliso Canyon is a depleted oil-field reservoir, not a salt cavern, and the instrument reads it as the endurance storage of §2.2. The field's loss reached gas-fired generation in the Los Angeles Basin, which lies in the electric sub-sector, and the community beside it, which lies in Healthcare and Public Health (Section

5.2). The Foundational Definitions are cited by designation, FD-REG holding their versions; FD-LD (*Layer Decomposition and Determinism*) bears on Section 6.8.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the root-cause analysis of the SS-25 failure by Blade Energy Partners, commissioned by the Commission and the state's Division of Oil, Gas and Geothermal Resources (now the Geologic Energy Management Division) and published on 16 and 17 May 2019; the Division's orders, its news release of 18 February 2016 and its account of the event; the Commission's record of the field's return to service and inventory limits; the Air Resources Board's estimate of the methane released; the Governor's proclamation of a state of emergency of 6 January 2016; the joint assessment and action plan of 5 April 2016 of the California Energy Commission, the Commission, the California Independent System Operator and the Los Angeles Department of Water and Power; and the report of the federal Interagency Task Force on Natural Gas Storage Safety of 18 October 2016. The Energy Information Administration's analyses of 9 August 2016 and 28 July 2017 give the field's capacity and its share of the company's storage. The company's releases are its own account. No non-public material is used.

The root-cause report itself could not be retrieved in this pass. Its findings are taken from the Division's summary of it and from reporting of its release, and the instrument states only what those agree on (OI-3).

2.2 Where the Sources Differ

On when the leak was stopped and the well sealed, the dates depend on what is counted. The Division records a temporary seal on 11 February 2016, a permanent cement seal on 17 February, and its confirmation on 18 February that the well was no longer leaking; the Commission's timeline gives 12 February for the stop and 15 February for the seal. This instrument takes the Division's dates, the Division being the regulator that confirmed the seal, and dates the release as about sixteen weeks, the figure the Air Resources Board uses.

On the quantity released, the company estimated 4.62 billion cubic feet, about 84,200 metric tons, in May 2016, against the Board's preliminary estimate of 5.4 billion cubic feet; the Board's final estimate is 109,000 metric tons of methane. Each figure is its author's. No finding depends on the difference.

3. The Event

3.1 The Field

Aliso Canyon is a depleted oil field beside the Porter Ranch community of Los Angeles, converted to gas storage. Before the release it held up to 86 billion cubic feet of the company's

total storage capacity of 135 billion, and could send out up to 1.9 billion cubic feet a day of the 3.7 billion its four fields together could deliver (Energy Information Administration, 9 August 2016). It had 114 wells.

3.2 The Failure

The root-cause analysis found the direct cause to be a rupture of the well's 7-inch production casing, thinned by microbial corrosion from the outside where the casing was in contact with groundwater. Before the release, gas was injected and withdrawn through both the tubing and the space between the tubing and that casing, so that the casing was the only barrier between the stored gas and the formation around the well. The analysis found that the company had not investigated the causes of more than sixty casing leaks recorded at the field since the 1970s, and that it lacked a risk assessment focused on well integrity, systematic protection against external corrosion, and real-time continuous monitoring of well pressures (Division's summary). It also found that the company's practices since and the Division's post-leak regulations address most of the root causes it identified.

3.3 The Release and the Attempts to Stop It

The company detected the leak on 23 October 2015 and notified the Division. Top-kill operations began the next day, with a specialist well-control contractor engaged within two days; seven attempts to kill the well from the surface failed between 24 October and 22 December. A relief well, begun on 4 December, intersected SS-25 at a depth of about 8,500 feet, and mud and cement stopped the flow, with a temporary seal on 11 February 2016 and a permanent cement seal on 17 February. On 18 February the Division confirmed the well sealed, on temperature, noise and cement-bond tests developed with three national laboratories. The Air Resources Board estimates that 109,000 metric tons of methane were released over the sixteen weeks the well was uncontrolled, about 20 percent of the state's methane emissions over the period, and the federal task force called it the largest natural gas leak on record in the United States.

3.4 The Community

Residents reported headaches, nausea and dizziness, which the South Coast Air Quality Management District associated with the odorants added to the gas, stating that health experts did not expect long-term effects from them. The company paid for the temporary relocation of households that chose to leave: on 7 February 2016 it reported 4,505 households placed in temporary housing, 840 more being placed and 1,662 already checked out. On 6 January 2016 the Governor proclaimed a state of emergency, directing that withdrawals be maximized to reduce the field's pressure, that injection stay prohibited until an independent review of well safety was complete, that the company plan for a relief well's failure, and that emergency regulations apply to every storage operator in the state.

3.5 The Field's Loss to the System

The Division prohibited injection from 10 December 2015 (Order 1106), and the field was drawn down. After the leak the allowed inventory was 15 billion cubic feet, held for peak demand. On 5 April 2016 the four energy agencies found that, without gas from the field, the region faced up to fourteen days in the coming summer with gas shortages to electric power plants, and "a distinct possibility of electricity service interruptions"; they proposed eighteen mitigation measures. The sources read record that assessment and that plan; they do not record a curtailment of gas or an interruption of electricity in the winter of the release, and none is asserted here.

3.6 The Return to Service

On 19 July 2017 the Commission and the Division found the field safe to operate at reduced capacity. About 60 percent of its 114 wells had been taken out of operation and isolated; the rest had passed a battery of tests and been fitted with tubing and packers and real-time pressure monitoring. The maximum operating pressure was set at 2,926 pounds per square inch, equivalent to 68.6 billion cubic feet, 17.4 billion below the pre-leak capacity, and working gas was first capped at 23.6 billion cubic feet, about 28 percent of that capacity, with withdrawals allowed only when reliability required them. The Commission raised the cap to 41.2 billion cubic feet in November 2021 and to 68.6 billion on 31 August 2023. The field has not returned to its pre-leak capacity.

3.7 What Followed

The federal task force made 44 recommendations in October 2016, among them phasing out designs in which a single point of failure, as at SS-25, stands between the stored gas and the surface. The Pipeline and Hazardous Materials Safety Administration issued an interim final rule on 19 December 2016, responding to Section 12 of the PIPES Act of 2016 and incorporating by reference the American Petroleum Institute's recommended practices 1170 and 1171 for storage in salt caverns and in depleted reservoirs and aquifers. The Division adopted emergency regulations for every storage well in the state, as the proclamation directed.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-3) and effect elements (E-1 to E-5), as Table 4-1 sets out. Composed, they are the class's EF-1 at a depleted-reservoir field: a casing failure that the well's design made a release to the surface, a release that surface well control could not stop and a relief well stopped after sixteen weeks, a community moved away from it, and a field removed from the system by regulatory order and returned to it under caps that have not reached its pre-leak capacity.

Table 4-1. Event elements and their mapping. Section references are to *Underground Storage* v0.10.

No.	What happened	Class provision	Section
I-1	Production casing ruptured, thinned by external microbial corrosion from groundwater	Casing integrity failure (EF-1, §3.2)	3.2
I-2	Gas flowed through tubing and casing, the casing the only barrier; no real-time pressure monitoring; sixty earlier casing leaks not investigated	EF-1 adequacy standard; well-integrity monitoring (§3.2, §3.6)	3.2
I-3	Seven surface kills fail, 24 October to 22 December 2015; relief well begun 4 December	Emergency well control (§3.2); contractors (§7.2, §8.4)	3.3
E-1	Release for about sixteen weeks; 109,000 metric tons of methane on the Board's estimate	Uncontrolled release (EF-1); Tier 1 (§4.2)	3.3
E-2	Thousands of households relocated; symptoms reported; state of emergency proclaimed	No place in the taxonomy (§4.2); F-2	3.4
E-3	Injection prohibited and the field drawn down; about two-thirds of the company's storage capacity out	Endurance capacity (§2.2); Tiers 1 and 2 (§4.2)	3.1, 3.5
E-4	Up to fourteen summer days of assessed gas shortage to power plants; eighteen measures	XC-3 (§3.5); Tier 3 (§4.2)	3.5
E-5	Limited return on 19 July 2017 at a 23.6 billion cubic foot cap; caps of 41.2 (2021) and 68.6 (2023); never 86	Return-to-service hold (§4.4, §7.2, §8.3)	3.6

5. Tier Determination

5.1 The Four Characteristics

Intentionality is absent: the root-cause analysis attributes the failure to external corrosion, and no source read suggests that any party caused it. Persistence is absent as an adversary's; the release itself persisted for sixteen weeks, and its consequence for the field has persisted for years, but that is the hazard's and the regulator's, not an actor's. Precision is absent: the casing failed where groundwater had corroded it. Denial is present in effect, without an actor: the well's single barrier and the failure of seven surface kills denied control of the well for sixteen weeks, and what the class says an adversary would arrange, the denial of well control and of the field's return (§5.2), the failure and the regulatory hold supplied.

5.2 Determination

The event does not meet the threshold for the military-action extension. Under the author's rulings of 10 and 13 October 2026, recorded at FD-DBL (*Design Basis Layers*) §6.4, all four characteristics must be established, and three are absent. It is a non-adversarial storage well casing integrity failure and blowout realizing EF-1 in full (§3.2), at a depleted-reservoir field, with the cross-sector cascade of XC-3 assessed and planned against (§3.5). The provisional reading at conferral is confirmed. That is the tier determination returned for the DBA Reference Event Register. The impacted sectors proposed by the class are confirmed: Energy, electric sub-sector, for the gas shortages to power plants the energy agencies assessed in 2016, recorded under the author's ruling of 15 October 2026 that the register may name a sibling sub-sector; and Healthcare and Public Health, for the relocation of households and the symptoms reported beside the field.

5.3 The Constructed Analog

The class names Scenario A (§5.3), an adversarial well integrity failure at a major salt cavern field, as the event's deliberate analog. That scenario runs forward and is not developed here. Aliso Canyon is a depleted-reservoir field, and its figures, sixteen weeks to a relief-well kill and years of capped return, are figures for that geology; a salt cavern well's control time and return are not read from them. The bearing on Scenario A is recorded in F-1 and F-3 and offered to Project 8.

6. Characterization Against the Design Basis

6.1 Preface, §1.4, §3.2 and §5.3.3 The Class's Account of the Event

Held, with a gap. The account holds in direction: a corroded casing failed, the release was very large, the community was moved and the return to service took years. Five points were qualified and went to the class as Erratum 1 of Project 11's note of 1 November 2026; the class took in all five at its v0.10 of the same day, so no erratum remains owed, and they are recorded here as the reading that found them. The Preface and EF-1 call the release an eleven-week blowout; it lasted about sixteen weeks, from 23 October 2015 to the temporary seal of 11 February 2016, the well being confirmed sealed on 18 February. EF-1 says the event removed roughly 15 percent of Southern California working gas storage; the field held 86 of the company's 135 billion cubic feet, about two-thirds. The Preface speaks of supply curtailment during a winter heating period; the sources read record an injection ban, a drawdown, and an assessed risk of summer shortages with a plan against it, not a curtailment that winter. EF-1 says the release forced the evacuation of Porter Ranch; the households that left were relocated temporarily, by choice, at the company's expense. And §1.4, §4.1 and §5.3.3 give the return to service as nearly five years, to partial or to full operations; limited operations were authorized on 19 July 2017, about twenty-one months after the leak began, the cap reached 68.6 billion cubic feet on 31 August 2023, and the field has

not returned to its pre-leak 86. The figure of about 5 billion cubic feet released holds against the company's and the Board's estimates.

6.2 §3.2 EF-1 Storage Well Casing Integrity Failure and Blowout

Held, with a gap. The event is EF-1 as written: a casing failed from corrosion and produced an uncontrolled release. The class's adequacy standard is well-integrity monitoring that detects casing anomalies before progression, and emergency well control available within a defined response time. At SS-25 there was no real-time pressure monitoring, sixty earlier casing leaks had not been investigated, the casing was the only barrier because gas flowed through it, and the control that worked was a relief well, not the surface kill. Finding F-1.

6.3 §4.1 and §4.2 The Consequence Taxonomy

Held, with a gap. The class's three tiers are the field, the transmission system and the cross-sector cascade, all measured in deliverability and supply. The event's first consequence fell on neither: it fell on the households beside the field and on the air. The taxonomy has no place for it. Finding F-2.

6.4 §4.4, §7.2 and §8.3 The Return-to-Service Hold

Held, with a gap. The class takes the regulatory return-to-service process as the governing recovery constraint for a well-integrity event, and the event confirms it: the field's recovery was set by the regulators, not by repair. What the event adds is that the process did not end in a return. It ended in a derating, a pressure limit below pre-leak capacity, and a sequence of inventory caps set over six years on the region's need, within an investigation into whether to use the field at all. The review of every well before any return, which §7.2 attributes to the federal storage rule, appears in the record read as the state's requirement, under the Division's Order 1109 and the Governor's proclamation; this instrument has not read the federal rule's text for such a requirement and makes no finding on it. Finding F-3.

6.5 §2.2, §3.5 and §4.2 The Field's Share and the Cascade to Generation

Held, with a gap. The class treats a depleted-reservoir field as endurance capacity whose loss is offset in the short term, and names the cascade to gas-fired generation (XC-3) and the winter compound (XC-4). One field held about two-thirds of the region's storage and half its sendout, and the energy agencies found up to fourteen summer days of gas shortages to power plants without it. The field was removed not by damage to it but by an order to protect the public. Finding F-4.

6.6 §7.2 and §8.4 Blowout Control Contractors

Held. A specialist well-control contractor was engaged within two days, and one relief well stopped the release. A single event did not stress the pool of contractors the class is concerned

with; the sixteen weeks were set by the time to drill and intersect a relief well, which F-1 takes up.

6.7 §2.6 Regulatory Authority

Held. The class records the federal storage rule as a response to Aliso Canyon, and the record confirms it: the interim final rule of 19 December 2016 under Section 12 of the PIPES Act of 2016, incorporating the recommended practices for depleted reservoirs and salt caverns, and the task force's 44 recommendations. The class's further statement that the adequacy of those rules against the military-action tier has not been assessed is not tested by a non-adversarial event.

6.8 §3.6 The AI-Governance Design Basis

Not exercised. No machine-learning or generative system bore on the release. The class's requirement that well-integrity monitoring receive its data by paths independent of the control path presupposes that monitoring exists; at SS-25 the root-cause analysis found no real-time continuous pressure monitoring, which F-1 takes up as a design-basis matter rather than an AI-governance one. FD-LD (*Layer Decomposition and Determinism*) is not under load.

6.9 §5 The Military-Action Extension

Not exercised. No act of any party bore on the event (Section 5.1).

6.10 Summary of Design Basis Performance

Of the nine readings, five Held, with a gap, the class's account among them; two Held; and two were Not exercised. Nothing was Missed. EF-1 was written from this event and reads it. What the event adds is that the time to control a well is the time to a relief well where the casing is the only barrier, that the first consequence of a storage blowout falls on the people beside the field, that a return to service can be a permanent derating, and that one field's share of a region can make a non-adversarial loss a regional reliability event.

7. Findings

Section references are to *Underground Storage* v0.10 unless another instrument is named.

7.1 F-1 — Well Control Is Bounded by the Relief Well Where the Casing Is the Only Barrier

Gas flowed through the casing as well as the tubing, so the casing was the only barrier; no real-time pressure monitoring existed; more than sixty earlier casing leaks had not been investigated; seven surface kills failed over two months; and a relief well stopped the release after sixteen weeks (Sections 3.2 and 3.3). The finding confirms EF-1 and its adequacy standard, and is that the class should state, as part of EF-1, that a well whose production casing is its only barrier

between stored gas and the formation carries the event as a release to the surface rather than a contained failure; that the emergency well-control response time of the adequacy standard is bounded, for such a well, by the time to drill and intersect a relief well, not by the time to a surface kill, and is stated as such; that well-integrity monitoring is credited only where continuous pressure monitoring of the tubing and every annulus exists; and that each recorded casing leak at a field is investigated for cause and read across its other wells.

Affected: §3.2, §3.6, §7.2. Disposition: R-1.

7.2 F-2 — The First Consequence of a Storage Blowout Falls Beside the Field

Thousands of households left their homes for months, residents reported symptoms the air district associated with the gas's odorants, the Governor proclaimed a state of emergency, and the federal task force called the release the largest natural gas leak on record in the country (Sections 3.3 and 3.4). The class's consequence taxonomy measures the field, the transmission system and the cascade to generation (§4.2), and does not count a release that reaches the people beside the field. The finding is that the class should add to §4.2 a consequence chain for the release itself, measured by the population within reach of the release and the duration of their displacement, with the emergency planning for that population named as a dependency on the local authorities and on Healthcare and Public Health; and that a field beside a residential community states that population as part of its EF-1 consequence.

Affected: §3.2, §4.1, §4.2. Disposition: R-2.

7.3 F-3 — A Return to Service May Be a Permanent Derating

Limited operations were authorized twenty-one months after the leak began, with about 60 percent of the wells isolated, a pressure limit equivalent to 68.6 billion cubic feet against a pre-leak 86, and a working-gas cap of 23.6 billion that the Commission raised to 41.2 billion in 2021 and to 68.6 billion in 2023, within an investigation into the feasibility of minimizing or eliminating the field's use (Section 3.6). The finding confirms §7.2 and §8.3 in taking the regulatory process as the governing recovery constraint, and is that the class should state the outcome of that process for a well-integrity event as a sequence of authorized capacities rather than a single return date, each with its date and its basis; that the bounding case of §4.1 counts a permanent loss of part of the field's capacity, and the possibility that policy, not repair, ends its use; and that the emergency return-to-service framework of §8.3 distinguishes the wells involved in the event from the rest of the field, as the hold here, which kept every well out of service until it was tested or isolated, did not.

Affected: §1.4, §4.1, §4.4, §7.2, §8.3. Disposition: R-3.

7.4 F-4 — A Field's Share of Its Region Sets the Cascade, Whatever Removes It

The field held about two-thirds of the company's storage capacity and about half its sendout; its loss was imposed by an order prohibiting injection and by a drawdown the Governor directed to reduce its pressure; and the energy agencies found up to fourteen summer days of gas shortages to power plants without it (Sections 3.1, 3.4 and 3.5). The class treats depleted-reservoir storage as endurance capacity that the system can offset in the short term (§2.2), which holds for a region with many fields and does not for one in which a single field dominates. The finding confirms XC-3 and is that the class should state, for each field, its share of the storage capacity and deliverability of the region and of the generation it serves; that a field whose share exceeds a stated threshold carries XC-3 at the regional level even in the non-adversarial domain; and that the loss of a field by a protective order or a drawdown to reduce its pressure is a consequence path of EF-1 in its own right, to be planned against as the agencies did here.

Affected: §2.2, §3.5, §4.2. Disposition: R-4.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 to R-4 dispose to *Underground Storage*; R-4 is noted to the Natural Gas Sector Framework and to the electric sub-sector.

No.	Title	Affected	Recommendation	Pri.
R-1	Bound well control by the relief well	§3.2, §3.6, §7.2	Carry EF-1 as a release to surface for a well whose casing is its only barrier; bound its response time by the time to a relief well; credit monitoring only with continuous pressure monitoring of tubing and every annulus; investigate each casing leak and read it across the field.	1
R-2	Count the release beside the field	§3.2, §4.1, §4.2	Add a consequence chain for the release itself, measured by the population within its reach and the duration of displacement, naming emergency planning as a dependency on local authorities and Healthcare and Public Health.	1
R-3	State return to service as a sequence of capacities	§1.4, §4.1, §4.4, §7.2, §8.3	State the outcome as dated, authorized capacities; count permanent derating and policy closure in the bounding case; distinguish the event's wells from the rest	2

			of the field in the emergency framework.	
R-4	State each field's regional share	§2.2, §3.5, §4.2	State each field's share of regional storage, deliverability and the generation it serves; carry XC-3 regionally above a threshold; plan for loss by protective order or pressure-reducing drawdown as an EF-1 path.	2

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2 and the impacted sectors it confirms. OI-2: the class took in Erratum 1 at v0.10 and cites this instrument by designator; the findings, R-1 to R-4, are held for the author's discussion and are dispositioned at a later revision of the class, and of the framework and the electric sub-sector for R-4. OI-3: the root-cause report was read through the Division's summary and reporting of its release, not in its original; when it is read, any point on which it differs is recorded and the instrument revised. OI-4: the record is open as to the field's future, before the Commission; a decision to close the field, or to change its cap, is recorded at the next revision.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of fault.

10.2 Relationship to the Class's Scenarios and Use Case

The class's Scenario A (§5.3) transposes this event's mechanism to a salt cavern field; the figures here are for a depleted reservoir and should be carried to Scenario A only with that difference stated (Section 5.3). The series use case *Gas-Electric Cascade* (DBA-EN-UC-1) traces the gas-to-electric direction that XC-3 names; this event supplies its non-adversarial figures, one field holding about two-thirds of a region's storage and up to fourteen summer days of assessed shortage to power plants. F-3 and F-4 bear most on it.

10.3 Relationship to Other Instruments

Arkema Crosby Fire (DBA-CHEM-RE-2) reads a release that left a plant and moved a community, and its denial present in effect without an actor is the same reading as Section 5.1. Winter Storm Uri, the class's anchor event for NH-1, is not conferred; if it is, its reading of XC-4 should be taken with F-4.

References

Blade Energy Partners, *SS-25 Root Cause Analysis*, report to the California Public Utilities Commission and the Division of Oil, Gas and Geothermal Resources, May 2019, as summarized by the Geologic Energy Management Division. California Department of Conservation, Division of Oil, Gas and Geothermal Resources, News Release 2016-05, *State Regulators Confirm Aliso Canyon Natural Gas Well is Permanently Sealed*, 18 February 2016, and Orders 1106, 1109 and 1118. California Public Utilities Commission, Aliso Canyon well failure record, including the decision of 19 July 2017 and the inventory decision of 31 August 2023, and Investigation 17-02-002. California Air Resources Board, *Aliso Canyon leak emitted 109,000 metric tons of methane*, and its preliminary estimates. Office of the Governor, proclamation of a state of emergency, 6 January 2016. California Energy Commission, California Public Utilities Commission, California Independent System Operator and Los Angeles Department of Water and Power, *Aliso Canyon Action Plan to Preserve Gas and Electric Reliability for the Los Angeles Basin* and technical assessment, 5 April 2016. Interagency Task Force on Natural Gas Storage Safety, *Ensuring Safe and Reliable Underground Natural Gas Storage*, 18 October 2016. Pipeline and Hazardous Materials Safety Administration, interim final rule on underground natural gas storage, 19 December 2016. U.S. Energy Information Administration, *Today in Energy*, 9 August 2016, 28 July 2017 and 27 September 2023. South Coast Air Quality Management District, fact sheet on health impacts, revised 29 January 2016. Southern California Gas Company, incident update of 8 February 2016 and statement on the settlement with the Los Angeles County District Attorney, 13 September 2016. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 1 November 2026: issued in Project 11 under the Natural Gas Sector Framework v1.12, on the conferral of the framework's §10.4 of 15 October 2026, read against *Underground Storage* v0.9. Written new under the Reference Event Compression Standard v0.3, the energy series' first reference event instrument. Tier determination at Section 5.2, under FD-DBL (*Design Basis Layers*) §6.4 as ruled on 10 and 13 October 2026; F-1 to F-4, R-1 to R-4. The class's account holds in direction, with five qualifications going to the class as Erratum 1 of Project 11's note of 1 November 2026. Record open as to the field's future (OI-4).

Version 0.2, 1 November 2026: read against *Underground Storage* v0.10 and the Natural Gas Sector Framework v1.13, the versions Project 8 named for the CIKR example-site freeze, whose section numbers v0.9 shares. Section 6.1 records that the class took in Erratum 1 at v0.10, so no erratum is owed; Section 6.4 records that the review of every well before return appears in the record read as the state's requirement, on the point Project 8 raised; OI-2 is conformed. The tier determination, the characteristics, every finding and every recommendation are unchanged. Version 0.1 is in the project's _archive.